Design

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Input

For each hero:

Name

Base damage

Experience

For each wizard:

Max magic point For

each tank:

Armor level

Output

For each hero:

Level

Damage

Is alive

For each wizard:

Magic points

Max magic points

For each tank:

Armor

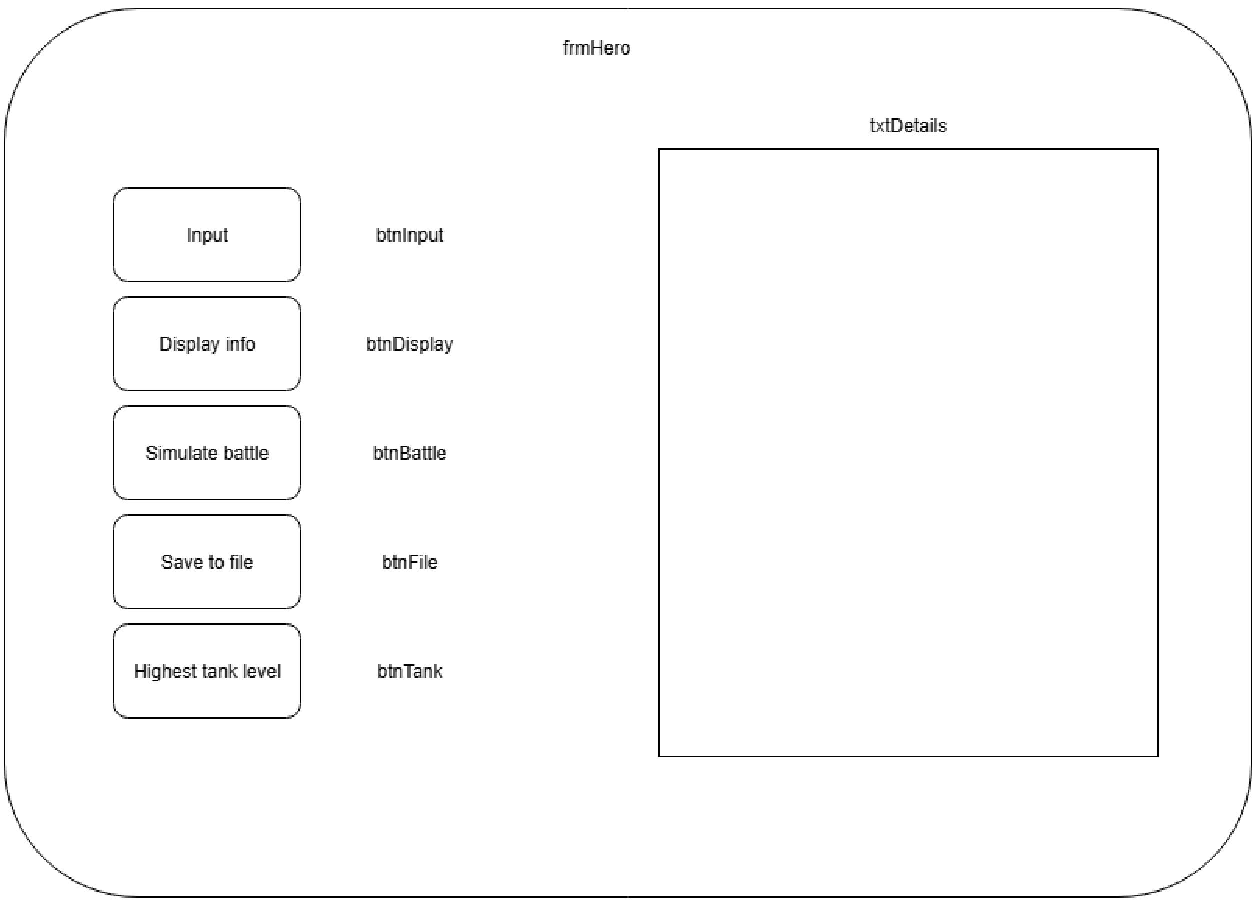
Defend

Events & Actions

Events	Actions

User clicks btnInput	Get number of heroes Resize arrays Get user input Instantiate and upcast oject
User clicks btnDisplay	Use polymorphism and display details
User clicks btnBattle	Choose 2 heroes Simulate battle
User clicks btnFile	Create file Save objects in file
User clicks btnTank	Check for tank with highest level

Interface

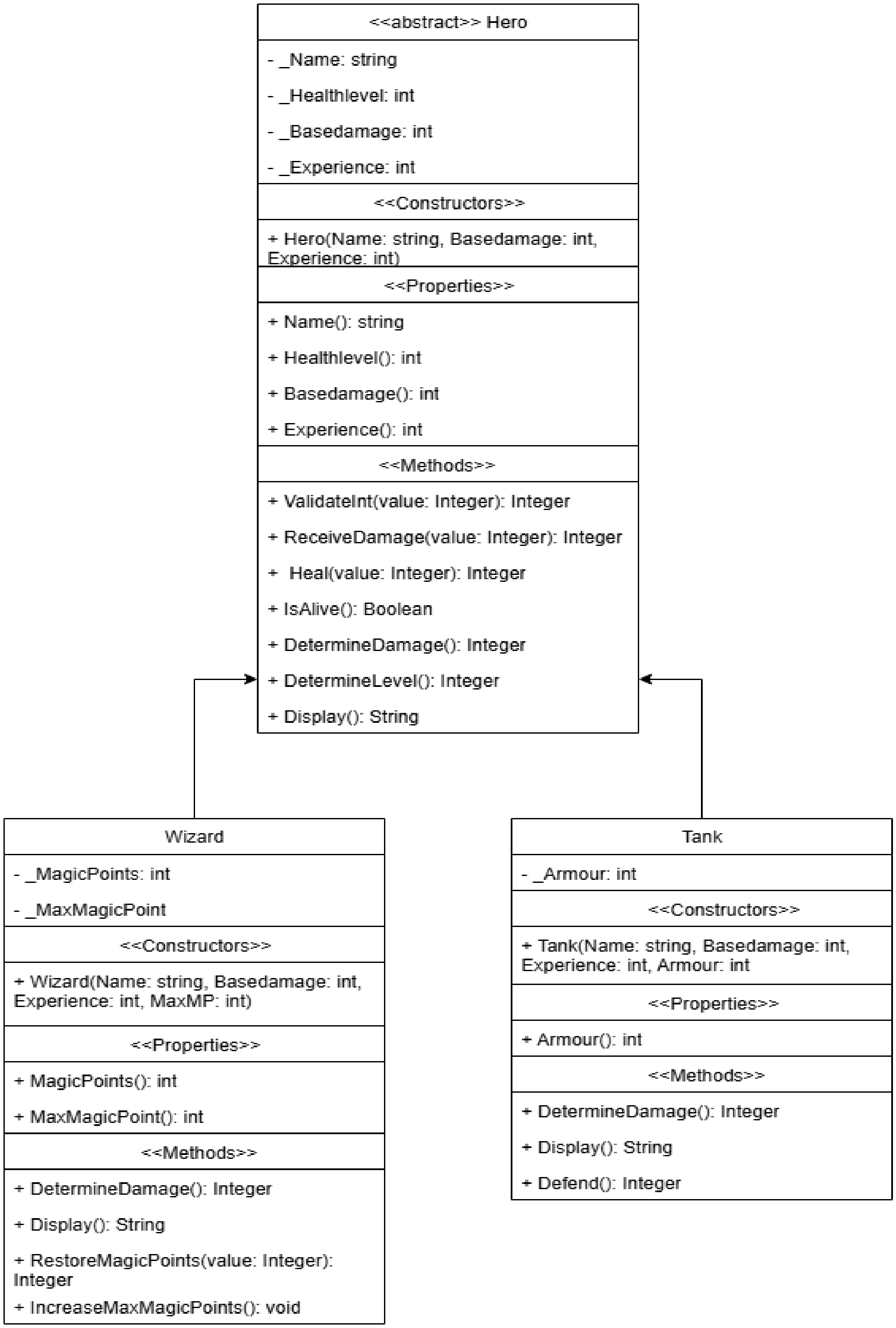


Variables

Private Numheroes: integer

Private heroes(): Hero

UML



<<abstract>> Hero
- _Name: string
- _Healthlevel: int
- _Basedamage: int
- _Experience: int
<<Constructors>>
+ Hero(Name: string, Basedamage: int, Experience: int)
<<Properties>>
+ Name(): string
+ Healthlevel(): int
+ Basedamage(): int
+ Experience(): int
<<Methods>>
+ ValidateInt(value: Integer): Integer
+ ReceiveDamage(value: Integer): Integer
+ Heal(value: Integer): Integer
+ IsAlive(): Boolean
+ DetermineDamage(): Integer
+ DetermineLevel(): Integer
+ Display(): String
Wizard
- _MagicPoints: int
- _MaxMagicPoint
<<Constructors>>
+ Wizard(Name: string, Basedamage: int, Experience: int, MaxMP: int)
<<Properties>>
+ MagicPoints(): int
+ MaxMagicPoint(): int
<<Methods>>
+ DetermineDamage(): Integer
+ Display(): String
+ RestoreMagicPoints(value: Integer): Integer
+ IncreaseMaxMagicPoints(): void
Tank
- _Armour: int
<<Constructors>>
+ Tank(Name: string, Basedamage: int, Experience: int, Armour: int
<<Properties>>
+ Armour(): int
<<Methods>>
+ DetermineDamage(): Integer
+ Display(): String
+ Defend(): Integer

Algorithm

1. NumHeroes <- Inputbox
2. Resize arrays
3. Get choice
4. Get name, base damage and experience
5. Get specific info for objects
6. Instantiate objects
7. Upcast to base class
8. TxtDetails.text <- Heroes(index).Display()
9. Enter name of 2 heroes
10. Find hero name and downcast
11. Simulate battle
12. Open and close file stream
13. Use binary formatter
14. Find highest level of tank

Test Data

Number heroes:	2	
Type:	Wizard	Tank
Name:	Merlin	Iron Tank
Health level:	100	100
Base damage:	45	70
Experience:	60	25
Level:	8	5
Damage:	309	175
Is alive:	Yes	Yes
Max magic point:	66	
Magic points:	66	
Armor:		33

Defend:		165